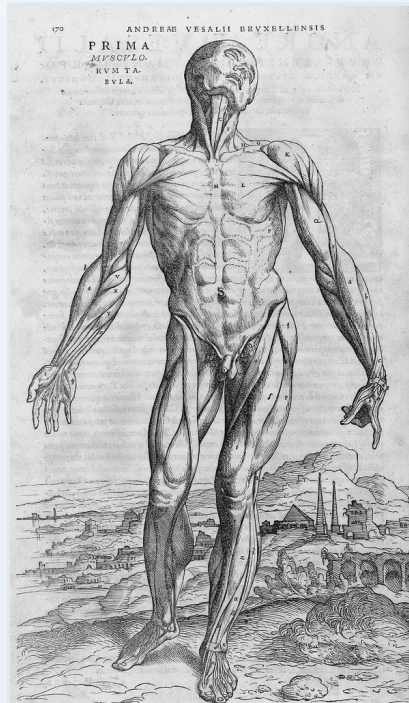


RENAISSANCE

Paper 1: Medicine Through Time with the Western Front



COURSE TEXTBOOK

Enquiry: How has medicine changed from c1250 to the modern day?

Contents

Lesson 1: KT2.1: The New Spirit of Enquiry: Humanism, The Printing Press & The Royal Society (c1500–c1700)

Lesson 2: KT2.2: Did treatments improve during the Renaissance?

Lesson 3: KT2.3: How significant were William Harvey and the Great Plague?

L1: KT2.1: The New Spirit of Enquiry: Humanism, The Printing Press & The Royal Society (c1500–c1700)

LEARNING OBJECTIVES

- ☐ Explain how the invention and spread of the printing press transformed the communication of medical ideas.
- ☐ Assess how the founding of the Royal Society and its motto 'Nullius in Verba' fostered empirical scientific enquiry.
- ☐ Evaluate why changes in communication did not immediately translate into better treatments or changing public beliefs about the causes of disease.

The Medical Renaissance was ignited by an intellectual rebellion against ancient dogma. Humanism encouraged scholars to believe their own eyes rather than blindly trusting 1,400-year-old manuscripts. When Gutenberg's printing press began churning out identical illustrated texts in the 1450s, and when the Royal Society received its royal charter in 1662 with the radical battle-cry 'Take nobody's word for it', the monopoly of the Catholic Church and Galen was permanently broken. Yet, behind this communication triumph lay a haunting historical paradox: revolutionary scientific networks did not discover a single new cure, leaving ordinary citizens clinging to the Four Humours and miasma for another two centuries.

Did you know?

- c.1440: Johannes Gutenberg invents the movable metal type printing press in Mainz, revolutionizing book production across Europe.
- 1662: King Charles II grants a Royal Charter to the Royal Society, establishing the motto Nullius in Verba ('Take nobody's word for it').
- 1665: Henry Oldenburg publishes Philosophical Transactions, the world's first peer-reviewed scientific journal.

Recall & Retrieval (Prior Knowledge: Medieval Medicine & The Black Death)

- | | |
|--|--|
| 1. In what year did the Black Death arrive in England? | 2. Who were the 'flagellants'? |
| 3. State one common medieval preventative measure for the Black Death. | 4. What are the four humours? |
| 5. What was 'miasma'? | 6. Which Roman physician developed the Theory of Opposites? |
| 7. How did medieval hospitals differ from modern ones? | 8. What was the purpose of 'bleeding' (phlebotomy) in medieval medicine? |
| 9. What was the role of a medieval apothecary? | 10. How did the medieval Church explain the cause of disease? |

Vocabulary Check

The Odd One Out: Circle *ONE* term that does not belong with the others. Explain your reasoning: What historical connection links the other terms, and why is your chosen term different?

Humanism | Movable Type | The Royal Society | Nullius in Verba | Philosophical Transactions | Empiricism | Animalcules | Miasma

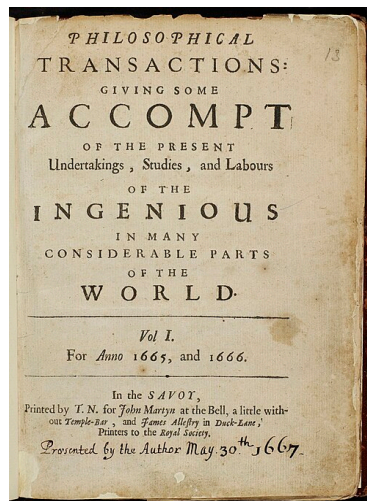
Teacher Model / Pupil Reasoning:

Source S1 : Renaissance Printing House with Movable Type (16th c.)



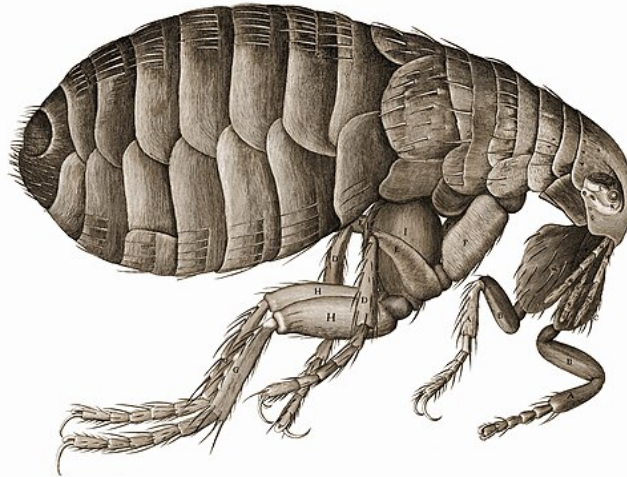
A Renaissance printing house showing compositors setting movable metal type and pressmen operating the heavy wooden screw press.

Source S2 : Title Page of Philosophical Transactions, Vol. I (1665–1666)



Frontispiece of Volume I of Philosophical Transactions (1665), the world's first peer-reviewed scientific journal, printed for John Martyn, printer to the Royal Society.

Source S3 : Robert Hooke's Engraving of a Flea from Micrographia (1665)



*Robert Hooke's famous fold-out engraving of a flea observed under the compound microscope, published by the Royal Society in *Micrographia* (1665).*

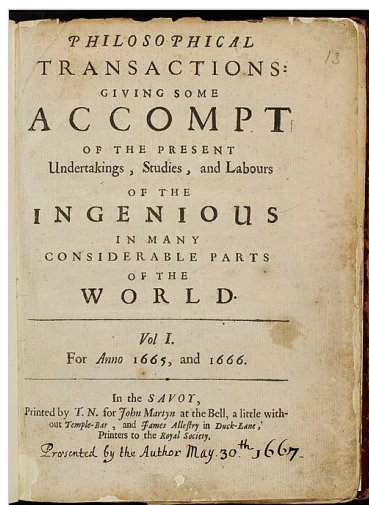
[1.1] The Medieval Stagnation: Why Ancient Dogma Refused to Die [1.1] For over a millennium, European medicine lived under the unquestioned authority of ancient Rome. Monks in Catholic monasteries painstakingly hand-copied the treatises of Galen and Hippocrates, treating their classical writings as infallible medical scripture. To challenge Galen was to risk excommunication or imprisonment. Medical students sat passively in university lecture halls while an elderly professor read aloud in Latin from an elevated pulpit (*cathedra*), while an uneducated barber-surgeon sliced an animal carcass on the floor below. No one dissected to discover anything new; dissection was merely a visual demonstration to prove that Galen was always right. ### [1.2] The Humanist Awakening: Believing Your Own Eyes Over Ancient Dust [1.2] By 1500, this intellectual stagnation began to fracture under the force of the **Renaissance**—a cultural and intellectual rebirth driven by **Humanism**. Humanist scholars in Italy and northern Europe sought out original Greek and Roman manuscripts that had not been distorted by centuries of monastic transcription. More importantly, humanism championed human reason, secular inquiry, and the moral duty to directly interrogate the physical world rather than blindly accepting traditional dogma. Scholars began to argue that knowledge should be derived from what human eyes could observe and verify. Yet, in the squalid, smoke-choked streets of Tudor and Stuart London, ordinary citizens remained untouched by university philosophy. When epidemic fevers struck, they still looked to the heavens in terror, convinced that disease was God's punishment for sin or the result of poisonous atmospheric vapors (*miasma*).



Source A: Source A: Early Gutenberg-Style Movable Type Printing Press (16th c.)

[2.1] Smashing the Scribes: Gutenberg's Machine Defies Church Censorship [2.1] The invention of the movable metal type printing press by **Johannes Gutenberg** in Mainz, Germany (c.1440), brought to England by **William Caxton** in 1476, transformed European communication. Prior to printing, every single medical manuscript had to be hand-copied by monastic scribes. This process took months per volume, cost vast sums of money, and routinely introduced compounding transcription errors and artistic distortions. Crucially, the Catholic Church controlled the scriptoria, enabling it to censor and suppress any text that challenged Galenic or theological orthodoxy. Gutenberg's press smashed this clerical bottleneck. Movable metal type allowed thousands of identical copies of a single text to be produced in days at a fraction of the cost, making censorship virtually impossible as books crossed borders faster than authorities could confiscate them. Furthermore, printers could reproduce complex, mathematically precise anatomical woodcuts identically, ensuring medical students in London, Paris, and Padua studied the exact same diagrams without error. ### [2.2] The Social Limits of Print: Latin Luxury vs Popular Superstition [2.2] Yet, this communications revolution had profound social and economic limitations. The overwhelming majority of pioneering medical treatises were printed in **Latin**, the international language of scholars, rendering them completely inaccessible to ordinary, uneducated people. Books remained expensive luxury items affordable only by wealthy physicians, nobles, and university libraries. For the ordinary peasant or apprentice in Tudor England, formal medical texts were irrelevant. Commercial printshops quickly realized that printing cheap astrological almanacs, folk herbals, and traditional Galenic health guides in English was far more lucrative than publishing dense scientific treatises. Consequently, rather than eradicating medieval superstitions, early commercial printing actually helped entrench traditional beliefs in humoral balance and planetary influences among the wider public.

Q1. GCSE Analytical Assessment: Evaluate both the revolutionary communication impact of the printing press and its historical limitations.

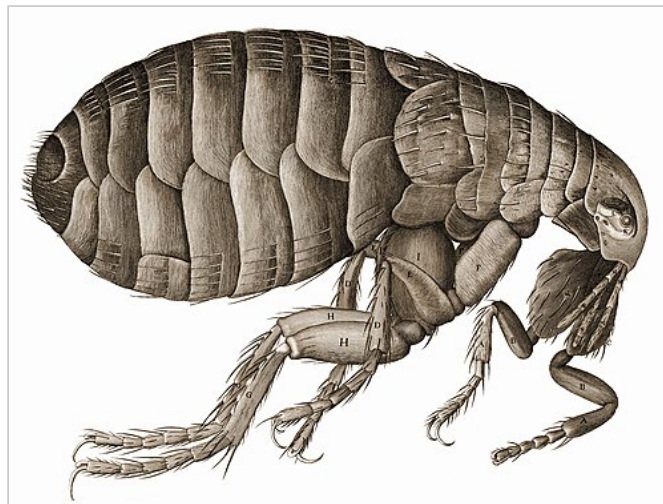


Source B: Source B: Title Page of *Philosophical Transactions*, Vol. I (1665–1666)

[3.1] 'Take Nobody's Word For It!': The Royal Charter of 1662 [3.1] By the mid-17th century, the humanist call for empirical enquiry had gained institutional power in Britain. In November 1660, a group of twelve natural philosophers met at Gresham College in London following a lecture by the young astronomer Christopher Wren. In 1662, King Charles II granted them an official royal charter, establishing **The Royal Society of London for Improving Natural Knowledge**. The Society adopted the radical Latin motto **Nullius in Verba** ("Take nobody's word for it"). They banned theological debate and declared that no scientific claim could be accepted as truth without physical experiment, mathematical calculation, or observable demonstration. The Royal Charter gave experimental science the official blessing and prestige of the English Crown, transforming natural philosophy from a suspicious, heretical pursuit into a prestigious national endeavor. ### [3.2] *Philosophical Transactions* (1665): The World's First Peer-Reviewed Journal [3.2] In 1665, the Royal Society's first secretary, Henry Oldenburg, created Europe's first peer-reviewed scientific journal: ***Philosophical Transactions*** (*Source B*). For the first time in medical

history, scientists had an established international network for sharing verified laboratory experiments, case notes, and observations. Instead of individual scholars keeping discoveries secret or relying on private letters, physicians across Europe could read, critique, and replicate each other's experiments. Oldenburg established the practice of peer review, ensuring that submitted papers were examined and verified by other members before publication, permanently raising the standards of scientific evidence. ### [3.3] The Optical Marvels: Robert Hooke & Antonie van Leeuwenhoek [3.3] The power of the new scientific method was demonstrated through optical technology. In 1665, Royal Society curator **Robert Hooke** published *Micrographia*, a breathtaking folio volume featuring intricate, copperplate engravings of everyday objects seen through the compound microscope. His giant, fold-out illustration of a common flea (*Source C*) stunned the public, revealing complex joints, bristles, and claws never before seen by human eyes. A decade later, an uneducated Dutch draper named **Antonie van Leeuwenhoek** used simple, high-power single-lens microscopes to observe scrapings from his own teeth, discovering millions of microscopic organisms swimming in water—creatures he named 'animalcules' (bacteria and protozoa). When Leeuwenhoek submitted his observations to the Royal Society in 1676 and 1683, the Society verified his findings, confirming that a previously invisible living universe existed.

Q2. GCSE Analytical Assessment: Evaluate both the institutional breakthrough of the Royal Society and its practical medical limitations.



Source C: *Source C: Robert Hooke's Fold-Out Engraving of a Flea from Micrographia (1665)*

[4.1] Brilliant Minds, Helpless Patients: The Great Renaissance Paradox [4.1] Despite the breathtaking achievements of Gutenberg's press, the Royal Society, and microscope technology, the Renaissance revealed a tragic historical paradox: **revolutionary scientific communication did not cure disease**. While Hooke and Leeuwenhoek revealed microscopic structures and 'animalcules', no 17th-century scientist understood what these tiny organisms were or what they did. Because the chemical and biological mechanisms of infection were completely unknown, 'animalcules' were treated merely as fascinating optical curiosities and parlor tricks rather than deadly pathogens. The true cause of infectious illness—microscopic germs—remained entirely undiscovered for another two centuries until Louis Pasteur published his Germ Theory in 1861. ### [4.2] Everyday Continuity at the Bedside: Leeches, Miasma and Astrological Almanacs [4.2] Consequently, for the vast majority of sick patients in Britain between 1500 and 1700, medical treatment experienced almost complete continuity with the Middle Ages. Knowing that microscopes could magnify a flea (*Source C*) or that the Royal Society debated air pressure did not save a single child dying of smallpox or cholera. When fevers struck, licensed university physicians still reached for their lancets to perform phlebotomy (bleeding) or prescribed violent purges to balance the Four Humours. Ordinary families continued to rely on village wise women, local apothecaries, and herbal remedies. The scientific revolution had permanently transformed how medical ideas were investigated, verified, and communicated among the educated elite; but inside the sickroom, ancient Galenic and humoral practices held sway.

Pair & Share Activity

Q3. undefined

GCSE Exam Practice

L2: KT2.2: Did treatments improve during the Renaissance?

LEARNING OBJECTIVES

- ☐ Analyze the continuities and changes in Renaissance treatments and preventions, including the impact of exploration and alchemy.
- ☐ Explain how institutional changes, specifically the Dissolution of the Monasteries, altered hospital care in England.

The Medical Renaissance saw an explosion of scientific curiosity, yet this rarely translated into better everyday medical care. While pioneer anatomists fundamentally transformed medical training by proving ancient authorities wrong, the average patient still relied on traditional herbal remedies, bleeding, and local wise women. The unexpected closure of monastery hospitals forced a massive rethink in healthcare, slowly paving the way for specialised hospitals focused on actually curing the sick.

Did you know?

- 1536: The year King Henry VIII initiated the Dissolution of the Monasteries, forcing the vast majority of English hospitals to close.
- 1543: The exact year Andreas Vesalius published his revolutionary, highly accurate anatomy textbook, *On the Fabric of the Human Body*.
- 300: The approximate number of anatomical mistakes Vesalius proved Galen had made because the Roman physician had dissected animals instead of humans.

Recall & Retrieval

- | | |
|---|---|
| 1. What was the Renaissance? | 2. What invention helped spread new medical ideas during the Renaissance? |
| 3. Who was Thomas Sydenham? | 4. Who wrote 'On the Fabric of the Human Body' (1543)? |
| 5. What did Vesalius prove about Galen's anatomical ideas? | 6. In what year did the Black Death arrive in England? |
| 7. Explain the role of the apothecary in medieval medicine. | 8. Why did the influence of the Church begin to decline in the Renaissance? |
| 9. What was the Royal Society? | 10. What was 'miasma'? |

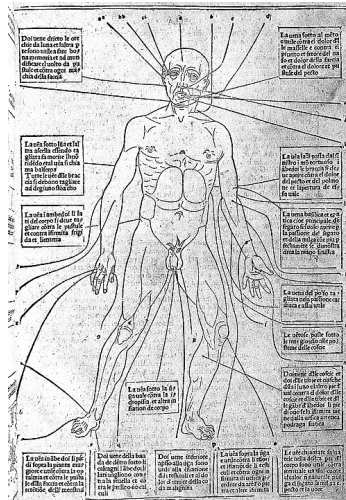
Vocabulary Check

The Golden Sentence: Choose TWO terms from the word bank. Write ONE grammatically sophisticated, historically accurate sentence connecting them using a causal conjunction (because, although, or consequently):

Transference | Iatrochemistry | Thomas Sydenham | Pharmacopoeia | Dissolution of the Monasteries
| Pest House

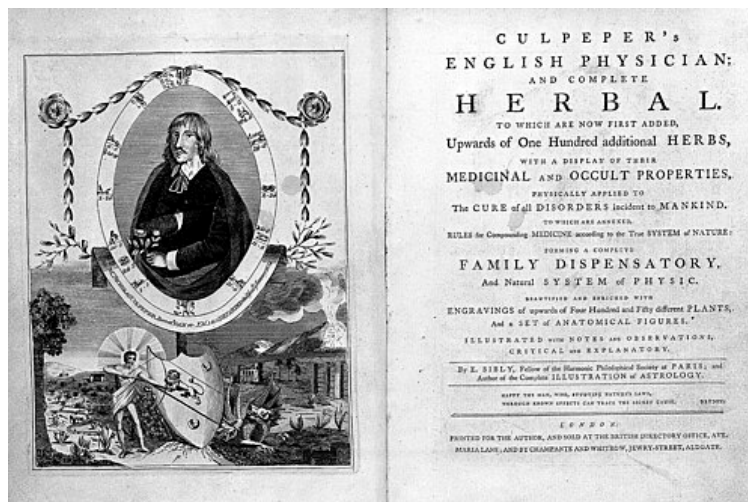
Teacher Model / Pupil Sentence:

Source S1 : 16th-Century Phlebotomy and Humoural Bloodletting Woodcut



Woodcut diagram from a 16th-century Italian medical text detailing specific vein points on the human body to be lanced for humoral bloodletting.

Source S2 : Nicholas Culpeper's English Physitian & Complete Herbal (1652)



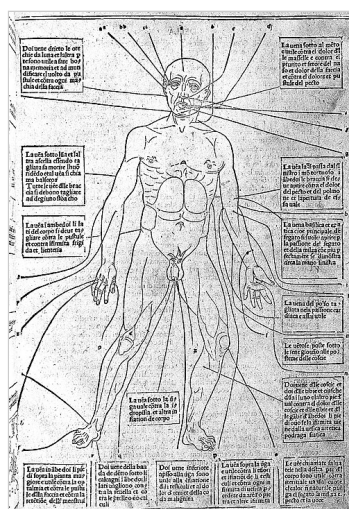
Title page of Nicholas Culpeper's *The English Physitian*, the first major medical guide to publish herbal remedies and medical knowledge in plain English.

Source S3 : Jan Steen: The Quack Doctor Selling Elixirs (c.1650)



Jan Steen satirical painting De Kwakzalver (The Quack Doctor), depicting an itinerant charlatan peddling miracle elixirs and teeth extractions on an outdoor stage.

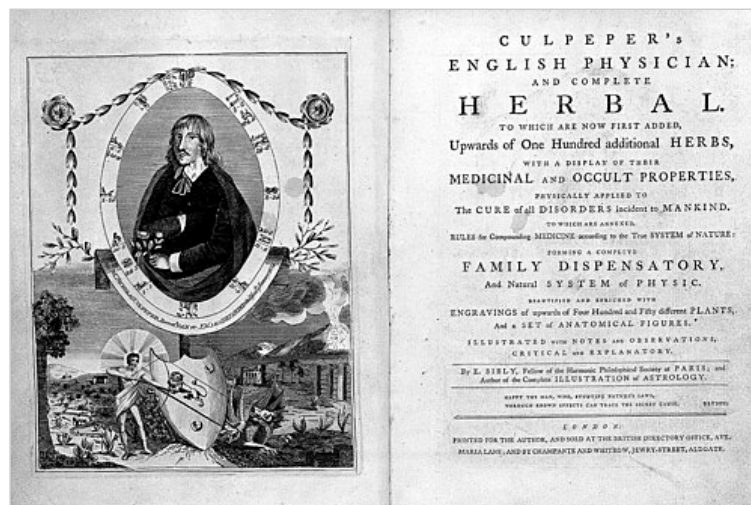
Treatments in the Renaissance Despite the brilliant discoveries in anatomy during the Renaissance, treatments for ordinary people hardly changed at all. Doctors still could not cure infectious diseases, so they continued to rely on traditional methods like bloodletting, purging, and herbal remedies. The only real changes in treatment came from the discovery of the 'New World' (the Americas). Explorers brought back new herbs and plants, such as the bark of the Cinchona tree, which was used to treat malaria, and sarsaparilla, which was used to treat syphilis. Furthermore, Thomas Sydenham, known as the 'English Hippocrates', began encouraging doctors to observe their patients' symptoms closely and treat the specific disease, rather than just treating the symptoms individually.



Source A: Source A: 16th-Century Phlebotomy and Humoural Bloodletting Woodcut

Continuity and Change in Treatment Bleeding, purging, and sweating remained the most common everyday treatments because the general public and physicians still firmly believed that an imbalance of the Four Humours caused disease. A newly popular theory was transference-the belief that an illness could be transferred to an object or animal (for example, rubbing warts with an onion, or sleeping next to a sheep to cure a fever). Global exploration brought new exotic herbal remedies from the New World (the Americas), such as the use of cinchona bark, which contains quinine, to successfully treat malaria. The growth of alchemy led to iatrochemistry (medical chemistry), popularized by the physician Paracelsus. This introduced powerful mineral-based cures, such as antimony and mercury, to forcefully purge the body of disease, which challenged traditional plant-based herbalism. Although exotic new herbs and powerful chemicals were introduced, treatment experienced massive continuity overall because doctors were still fundamentally trying to fix the body's internal balance rather than fighting germs.

Q1. GCSE Analytical Assessment: Evaluate the extent to which medical treatments improved during the Renaissance (c1500–c1700).



Source B: Source B: Nicholas Culpeper's English Physitian & Complete Herbal (1652)

Continuity and Change in Prevention People still desperately tried to avoid breathing in miasma (bad air) by carrying sweet-smelling herbs, and town councils continued to enforce the removal of rotting sewage from streets. People still followed the Regimen Sanitatis (moderation in diet and exercise), but also began using new instruments like barometers to try and link specific weather conditions to outbreaks of disease. Crucially, bathing became far less fashionable. The terrifying spread of syphilis (the Great Pox) across Europe had been linked to public bathhouses, so people instead tried to keep clean by regularly changing their linen clothes. Despite changing attitudes towards public bathing and a more scientific approach to observing the weather, prevention methods remained largely ineffective because the true microscopic causes of disease remained invisible.

KNOWLEDGE CHECK (Q2)

Why was the importation of cinchona bark from South America a more successful treatment than traditional humoral methods?

- ☐ A. It contained quinine, which actually treated the specific symptoms of malaria, rather than just attempting to rebalance humours.
- ☐ B. It was a holy relic blessed by the Pope.
- ☐ C. It killed the germs that caused malaria.
- ☐ D. It was used to draw bad humours out of the skin.

Q3. Describe the approach of Thomas Sydenham.



Source C: Source C: Jan Steen: *The Quack Doctor Selling Elixirs* (c.1650)

Medical Care: Hospitals and the Community In 1536, King Henry VIII ordered the Dissolution of the Monasteries. Because the vast majority of hospitals were run by monks and nuns on church land, the hospital system was devastated, shutting down almost all hospital beds in England. A few hospitals, such as St Bartholomew's, only survived because the citizens of London petitioned the King directly to refound it as a municipal hospital. Slowly, new charity-funded hospitals opened that actually employed physicians to focus on curing the sick, rather than just caring for their souls. Pest houses (or plague houses) began to appear; these were specialized isolation hospitals created specifically to quarantine patients suffering from highly contagious diseases. Apothecaries and surgeons also became more organised and professional during this period. In 1540, the Company of Barber-Surgeons was formed when Henry VIII merged the barbers' and surgeons' guilds, giving surgery its own formal training structure and separating it from simple haircutting for the first time. In 1617, apothecaries broke away from the Grocers' Company to form their own Society of Apothecaries, which trained and licensed its members to legally prepare and sell medicines, helping to squeeze out untrained 'quack' healers. Despite these changes, the vast majority of sick people continued to be cared for at home by female family members mixing cheap herbal remedies. The closure of the monasteries temporarily devastated the hospital system, but ultimately triggered a vital shift towards hospitals becoming places of practical medical treatment rather than religious rest homes.

Q4. Explain why treatments for ordinary people hardly changed during the Renaissance.

Q5. Describe one way apothecaries or surgeons became more professional during the Renaissance.

Medical Training and the Work of Vesalius Physicians still trained at universities, but dissection was finally legalised and became a central, highly fashionable part of their education. Students learned using highly detailed printed fugitive sheets (anatomical drawings with liftable paper layers). The most famous anatomist of the period was Andreas Vesalius, a lecturer in surgery at the University of Padua. In 1543, Vesalius published his groundbreaking, heavily illustrated anatomy book: *On the Fabric of the Human Body*. By performing dissections on executed criminals, Vesalius proved that Galen had made over 300 mistakes. For example, he proved the human jawbone is one solid bone, not two; that the breastbone has three segments, not seven; and that blood does not flow through invisible pores in the heart's septum. Vesalius revolutionized medical training because he proved that the ancient texts were flawed, actively encouraging generations of doctors to perform their own dissections and believe their own eyes rather than blindly trusting tradition.

Q6. To what extent did the 'New World' improve medical treatments?

Pair & Share Activity

Q7. In the period c1500-c1700, did the introduction of global herbal remedies and chemical cures represent a more significant improvement in treatment than the gradual reforms made to the training of physicians?

GCSE Exam Practice

Q8. Explain why there was little change in medical treatments during the Renaissance period (c1500-c1700), despite new anatomical discoveries. (12 marks)

Q9. Explain one way in which medical treatments in the Renaissance period (c1500-c1700) were similar to medical treatments in the medieval period (c1250-c1500). (4 marks)

L3: KT2.3: How significant were William Harvey and the Great Plague?

LEARNING OBJECTIVES

- ☐ Explain the significance of William Harvey's discovery and the reasons for its limited short-term impact.
- ☐ Compare the government and public responses to the 1665 Great Plague with the 1348 Black Death.

By the 17th century, medical knowledge was advancing, but practical treatment lagged far behind. William Harvey's brilliant discovery of blood circulation proved ancient theories wrong and laid the foundations for modern physiology, though it saved no lives at the time. Meanwhile, the devastating Great Plague of 1665 demonstrated that without an understanding of germs, even organized government action was powerless to stop a catastrophic epidemic.

Did you know?

- 1628: The year William Harvey published his groundbreaking book proving the circulation of the blood.
- 100,000: The approximate number of Londoners who died during the Great Plague of 1665 (one-fifth of the city's population).
- 28: The number of days an infected family was locked inside their house under the strict new quarantine rules.

Recall & Retrieval

- | | |
|---|--|
| 1. Who wrote 'On the Fabric of the Human Body' (1543)? | 2. What did Vesalius prove about Galen's anatomical ideas? |
| 3. What was 'transference'? | 4. What was the Royal Society? |
| 5. Why did the influence of the Church begin to decline in the Renaissance? | 6. In what year did the Great Plague hit London? |
| 7. How did local mayors try to prevent the spread of the Black Death in 1348? | 8. What was the Theory of the Four Humours? |
| 9. What was the purpose of 'bleeding' (phlebotomy) in medieval medicine? | 10. What was the role of a medieval apothecary? |

Vocabulary Check

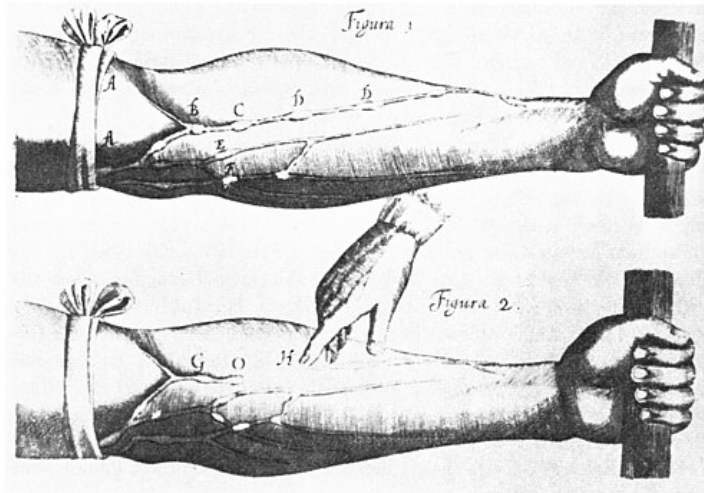
Conceptual Classification: Categorise the terms from the word bank into the two historical boxes below:

Circulation | William Harvey | Great Plague | Bills of Mortality | Searchers | Plague Doctor

Power, Governance & Warfare

Economy, Trade & Society

Source S1 : William Harvey's Arm Experiment with Valves in Veins (1628)



Engravings from Harvey's De Motu Cordis demonstrating how one-way valves in the arm veins permit blood to flow only toward the heart.

Source S2 : The London Bill of Mortality for the Great Plague (1665)

[illegible]

General Bill of Mortality for the City of London recording 7,165 plague deaths in a single week at the height of the epidemic in September 1665.

Source S3 : Engraving of 'Doktor Schnabel von Rom' (The Plague Doctor, 1656)



The iconic plague doctor costume: heavy waxed leather coat, spectacles, a long bird-like beak filled with aromatics, and a wooden cane to examine corpses.

William Harvey and the Great Plague In 1628, William Harvey made one of the most important discoveries in medical history: he proved that blood circulates around the body, pumped by the heart. He completely destroyed Galen's theory that blood was constantly being made in the liver and burned up by the muscles. Harvey proved this by dissecting cold-blooded animals and experimenting on human veins. However, in 1665, the Great Plague struck London, proving that while anatomical knowledge had improved, prevention of disease had not. People still blamed miasma and the planets, though local governments did take stronger action this time. The Mayor of London ordered victims to be locked in their houses with a red cross painted on the door, and banned public gatherings. Yet, just like in 1348, the true cause (fleas on rats) remained unknown.

William Harvey — Royal Physician

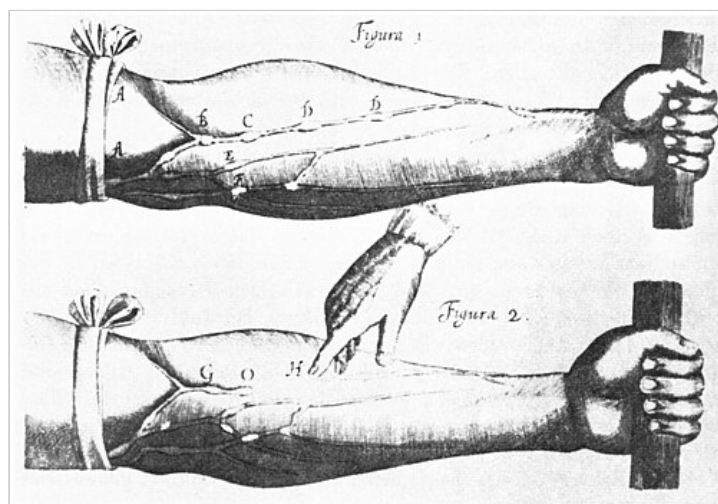
An English physician who discovered the full circulation of blood in the human body. By dissecting cold-blooded animals and experimenting with ligatures, he proved the heart acts as a pump, disproving Galen's theory that blood was constantly manufactured in the liver.

Key actions: Dissected cold-blooded animals to observe slow heartbeats, and used a system of rods and ligatures on human arms to prove that blood flows only one way through veins.

Significance: Published *De Motu Cordis* (1628). Proved that the heart acts as a muscular pump circulating blood around the body, disproving Galen's theory that blood was constantly manufactured by the liver and absorbed by the tissues.

Limitations: His discovery had no direct clinical treatment value at the time, as knowing blood circulated could not cure disease. He lacked microscopes powerful enough to prove the existence of capillaries (which connect arteries and veins; discovered by Malpighi in 1661). Many contemporary doctors rejected his findings, and his ideas did not appear in English university textbooks until decades later.

Q1. GCSE Analytical Assessment: Evaluate both the revolutionary physiology of William Harvey and its total lack of practical treatment impact.



Source A: Source A: William Harvey's Arm Experiment with Valves in Veins (1628)

William Harvey and the Circulation of the Blood William Harvey was an English doctor who studied at the famous medical university in Padua, Italy, before eventually becoming the personal royal physician to King Charles I. In 1628, he published his groundbreaking book, *An Anatomical Account of the Motion of*

the Heart and Blood in Animals. By using mechanical pump calculations and dissecting cold-blooded animals (whose slower heartbeats allowed him to observe muscle movement), he proved that blood circulates around the body in a one-way system, and that the heart acts as a muscular pump. He proved that the ancient Roman physician Galen was wrong when he claimed that blood was constantly manufactured in the liver and absorbed as fuel by the body. Harvey's work was a massive scientific breakthrough because it proved that traditional ancient texts were flawed and laid the vital foundations for modern physiology and future developments in surgery and blood transfusions.

Q2. Identify what William Harvey proved about the blood.

The Limits of Harvey's Discovery Despite his brilliance, Harvey's discovery had almost no immediate impact on everyday medical treatment because understanding how blood circulates did not help a doctor cure a fever or treat a disease in 1650. Many traditional doctors initially rejected his ideas because they went against Galen, and some even openly criticised him. It took until 1673 for his theories to be widely taught in universities. Furthermore, he could not explain how blood moved from arteries to veins because capillaries were invisible to the naked eye. This gap between brilliant new science and unchanged everyday treatment is a useful revision idea to remember (some teachers nickname it the 'Stagnation Paradox' as a memory aid, though it is not a term you need to quote in an exam): Vesalius and Harvey both had almost zero immediate impact on bedside treatment. Knowing that blood circulated did not give physicians new drugs, meaning patients still demanded traditional humoral bleeding and purging.

KNOWLEDGE CHECK (Q3)

What was the main reason why Harvey's discovery of blood circulation did not immediately change medical treatments?

- ☐ A. Because he refused to publish his findings in a book.
- ☐ B. Because the Church banned his book and executed him.
- ☐ C. Because knowing how the blood circulated did not provide doctors with any new drugs or cures for sick patients.
- ☐ D. Because he still believed the liver manufactured blood.

Q4. GCSE Analytical Assessment: Evaluate the advancements in government organization against the ongoing medical ignorance of the 1665 plague.

A general Bill for this present year, ending the 1st of December 1665, according to the Report made to the WISES men of London Weekly, By the Company of Parish Clerks of London, &c.

Parish	Male	Female	Total
St. Andrew's	10	12	22
St. Dunstons	15	18	33
St. Giles	20	25	45
St. Martin	25	30	55
St. Michael	30	35	65
St. Nicholas	35	40	75
St. Olave	40	45	85
St. Peter	45	50	95
St. Paul	50	55	105
St. Vincent	55	60	115
St. Andrew's	60	65	125
St. Dunstons	65	70	135
St. Giles	70	75	145
St. Martin	75	80	155
St. Michael	80	85	165
St. Nicholas	85	90	175
St. Olave	90	95	185
St. Peter	95	100	195
St. Paul	100	105	205
St. Vincent	105	110	215
St. Andrew's	110	115	225
St. Dunstons	115	120	235
St. Giles	120	125	245
St. Martin	125	130	255
St. Michael	130	135	265
St. Nicholas	135	140	275
St. Olave	140	145	285
St. Peter	145	150	295
St. Paul	150	155	305
St. Vincent	155	160	315
St. Andrew's	160	165	325
St. Dunstons	165	170	335
St. Giles	170	175	345
St. Martin	175	180	355
St. Michael	180	185	365
St. Nicholas	185	190	375
St. Olave	190	195	385
St. Peter	195	200	395
St. Paul	200	205	405
St. Vincent	205	210	415
St. Andrew's	210	215	425
St. Dunstons	215	220	435
St. Giles	220	225	445
St. Martin	225	230	455
St. Michael	230	235	465
St. Nicholas	235	240	475
St. Olave	240	245	485
St. Peter	245	250	495
St. Paul	250	255	505
St. Vincent	255	260	515
St. Andrew's	260	265	525
St. Dunstons	265	270	535
St. Giles	270	275	545
St. Martin	275	280	555
St. Michael	280	285	565
St. Nicholas	285	290	575
St. Olave	290	295	585
St. Peter	295	300	595
St. Paul	300	305	605
St. Vincent	305	310	615
St. Andrew's	310	315	625
St. Dunstons	315	320	635
St. Giles	320	325	645
St. Martin	325	330	655
St. Michael	330	335	665
St. Nicholas	335	340	675
St. Olave	340	345	685
St. Peter	345	350	695
St. Paul	350	355	705
St. Vincent	355	360	715
St. Andrew's	360	365	725
St. Dunstons	365	370	735
St. Giles	370	375	745
St. Martin	375	380	755
St. Michael	380	385	765
St. Nicholas	385	390	775
St. Olave	390	395	785
St. Peter	395	400	795
St. Paul	400	405	805
St. Vincent	405	410	815
St. Andrew's	410	415	825
St. Dunstons	415	420	835
St. Giles	420	425	845
St. Martin	425	430	855
St. Michael	430	435	865
St. Nicholas	435	440	875
St. Olave	440	445	885
St. Peter	445	450	895
St. Paul	450	455	905
St. Vincent	455	460	915
St. Andrew's	460	465	925
St. Dunstons	465	470	935
St. Giles	470	475	945
St. Martin	475	480	955
St. Michael	480	485	965
St. Nicholas	485	490	975
St. Olave	490	495	985
St. Peter	495	500	995
St. Paul	500	505	1005
St. Vincent	505	510	1015
St. Andrew's	510	515	1025
St. Dunstons	515	520	1035
St. Giles	520	525	1045
St. Martin	525	530	1055
St. Michael	530	535	1065
St. Nicholas	535	540	1075
St. Olave	540	545	1085
St. Peter	545	550	1095
St. Paul	550	555	1105
St. Vincent	555	560	1115
St. Andrew's	560	565	1125
St. Dunstons	565	570	1135
St. Giles	570	575	1145
St. Martin	575	580	1155
St. Michael	580	585	1165
St. Nicholas	585	590	1175
St. Olave	590	595	1185
St. Peter	595	600	1195
St. Paul	600	605	1205
St. Vincent	605	610	1215
St. Andrew's	610	615	1225
St. Dunstons	615	620	1235
St. Giles	620	625	1245
St. Martin	625	630	1255
St. Michael	630	635	1265
St. Nicholas	635	640	1275
St. Olave	640	645	1285
St. Peter	645	650	1295
St. Paul	650	655	1305
St. Vincent	655	660	1315
St. Andrew's	660	665	1325
St. Dunstons	665	670	1335
St. Giles	670	675	1345
St. Martin	675	680	1355
St. Michael	680	685	1365
St. Nicholas	685	690	1375
St. Olave	690	695	1385
St. Peter	695	700	1395
St. Paul	700	705	1405
St. Vincent	705	710	1415
St. Andrew's	710	715	1425
St. Dunstons	715	720	1435
St. Giles	720	725	1445
St. Martin	725	730	1455
St. Michael	730	735	1465
St. Nicholas	735	740	1475
St. Olave	740	745	1485
St. Peter	745	750	1495
St. Paul	750	755	1505
St. Vincent	755	760	1515
St. Andrew's	760	765	1525
St. Dunstons	765	770	1535
St. Giles	770	775	1545
St. Martin	775	780	1555
St. Michael	780	785	1565
St. Nicholas	785	790	1575
St. Olave	790	795	1585
St. Peter	795	800	1595
St. Paul	800	805	1605
St. Vincent	805	810	1615
St. Andrew's	810	815	1625
St. Dunstons	815	820	1635
St. Giles	820	825	1645
St. Martin	825	830	1655
St. Michael	830	835	1665
St. Nicholas	835	840	1675
St. Olave	840	845	1685
St. Peter	845	850	1695
St. Paul	850	855	1705
St. Vincent	855	860	1715
St. Andrew's	860	865	1725
St. Dunstons	865	870	1735
St. Giles	870	875	1745
St. Martin	875	880	1755
St. Michael	880	885	1765
St. Nicholas	885	890	1775
St. Olave	890	895	1785
St. Peter	895	900	1795
St. Paul	900	905	1805
St. Vincent	905	910	1815
St. Andrew's	910	915	1825
St. Dunstons	915	920	1835
St. Giles	920	925	1845
St. Martin	925	930	1855
St. Michael	930	935	1865
St. Nicholas	935	940	1875
St. Olave	940	945	1885
St. Peter	945	950	1895
St. Paul	950	955	1905
St. Vincent	955	960	1915
St. Andrew's	960	965	1925
St. Dunstons	965	970	1935
St. Giles	970	975	1945
St. Martin	975	980	1955
St. Michael	980	985	1965
St. Nicholas	985	990	1975
St. Olave	990	995	1985
St. Peter	995	1000	1995
St. Paul	1000	1005	2005
St. Vincent	1005	1010	2015
St. Andrew's	1010	1015	2025
St. Dunstons	1015	1020	2035
St. Giles	1020	1025	2045
St. Martin	1025	1030	2055
St. Michael	1030	1035	2065
St. Nicholas	1035	1040	2075
St. Olave	1040	1045	2085
St. Peter	1045	1050	2095
St. Paul	1050	1055	2105
St. Vincent	1055	1060	2115
St. Andrew's	1060	1065	2125
St. Dunstons	1065	1070	2135
St. Giles	1070	1075	2145
St. Martin	1075	1080	2155
St. Michael	1080	1085	2165
St. Nicholas	1085	1090	2175
St. Olave	1090	1095	2185
St. Peter	1095	1100	2195
St. Paul	1100	1105	2205
St. Vincent	1105	1110	2215
St. Andrew's	1110	1115	2225
St. Dunstons	1115	1120	2235
St. Giles	1120	1125	2245
St. Martin	1125	1130	2255
St. Michael	1130	1135	2265
St. Nicholas	1135	1140	2275
St. Olave	1140	1145	2285
St. Peter	1145	1150	2295
St. Paul	1150	1155	2305
St. Vincent	1155	1160	2315
St. Andrew's	1160	1165	2325
St. Dunstons	1165	1170	2335
St. Giles	1170	1175	2345
St. Martin	1175	1180	2355
St. Michael	1180	1185	2365
St. Nicholas	1185	1190	2375
St. Olave	1190	1195	2385
St. Peter	1195	1200	2395
St. Paul	1200	1205	2405
St. Vincent	1205	1210	2415
St. Andrew's	1210	1215	2425
St. Dunstons	1215	1220	2435
St. Giles	1220	1225	2445
St. Martin	1225	1230	2455
St. Michael	1230	1235	2465
St. Nicholas	1235	1240	2475
St. Olave	1240	1245	2485
St. Peter	1245	1250	2495
St. Paul	1250	1255	2505
St. Vincent	1255	1260	2515
St. Andrew's	1260	1265	2525
St. Dunstons	1265	1270	2535
St. Giles	1270	1275	2545
St. Martin	1275	1280	2555
St. Michael	1280	1285	2565
St. Nicholas	1285	1290	2575
St. Olave	1290	1295	2585
St. Peter	1295	1300	2595
St. Paul	1300	1305	2605
St. Vincent	1305	1310	2615
St. Andrew's	1310	1315	2625
St. Dunstons	1315	1320	2635
St. Giles	1320	1325	2645
St. Martin	1325	1330	2655
St. Michael	1330	1335	2665
St. Nicholas	1335	1340	2675
St. Olave	1340	1345	2685
St. Peter	1345	1350	2695
St. Paul	1350	1355	2705
St. Vincent	1355	1360	2715
St. Andrew's	1360	1365	2725
St. Dunstons	1365	1370	2735
St. Giles	1370	1375	2745
St. Martin	1375	1380	2755
St. Michael	1380	1385	2765
St. Nicholas	1385	1390	2775
St. Olave	1390	1395	2785
St. Peter	1395	1400	2795
St. Paul	1400	1405	2805
St. Vincent	1405	1410	2815
St. Andrew's	1410	1415	2825
St. Dunstons	1415	1420	2835
St. Giles	1420	1425	2845
St. Martin	1425	1430	2855
St. Michael	1430	1435	2865
St. Nicholas	1435	1440	2875
St. Olave	1440	1445	2885
St. Peter	1445	1450	2895
St. Paul	1450	1455	2905
St. Vincent	1455	1460	2915
St. Andrew's	1460	1465	2925
St. Dunstons	1465	1470	2935
St. Giles	1470	1475	2945
St. Martin	1475	1480	2955
St. Michael	1480	1485	2965
St. Nicholas	1485	1490	2975
St. Olave	1490	1495	2985
St. Peter	1495	1500	2995
St. Paul	1500	1505	3005
St. Vincent	1505	1510	3015
St. Andrew's	1510	1515	3025
St. Dunstons	1515	1520	3035
St. Giles	1520	1525	3045
St. Martin	1525	1530	3055
St. Michael	1530	1535	3065
St. Nicholas	1535	1540	3075
St. Olave	1540	1545	3085
St. Peter	1545	1550	3095
St. Paul	1550	1555	3105
St. Vincent	1555	1560	3115
St. Andrew's	1560	1565	3125
St. Dunstons	1565	1570	3135
St. Giles	1570	1575	3145
St. Martin	1575	1580	3155
St. Michael	1580	1585	3165
St. Nicholas	1585	1590	3175
St. Olave	159		

mistakenly believing they spread the disease. Although the government took a much more active and organized role in trying to prevent the plague, their efforts ultimately failed because their actions were based on incorrect theories about bad air rather than exterminating the true culprits: the rats and fleas.

Q6. Evaluate the short-term significance of Harvey's discovery.

Pair & Share Activity

Q7. Why was William Harvey's brilliant discovery of blood circulation completely ignored in the prevention and treatment of the Great Plague of 1665?

GCSE Exam Practice

Q8. Explain why the government's response to the Great Plague of 1665 was more effective than its response to the Black Death of 1348. (12 marks)

Q9. Explain one way in which the response to the Great Plague in London (1665) was different from the response to the Black Death (1348). (4 marks)